

ASSIGNMENT- 3

I. Check whether a number is Armstrong or not.

CODE:-

```
<?php
$number = readline("Enter A Number: ");
$a = $number;
$sum = 0;
while ($a != 0) {
    $rem = $a % 10;
    $sum = $sum + ($rem * $rem * $rem);
    $a = (int)($a / 10);
}
if ($number == $sum) {
    echo "Yes, $number is an Armstrong Number";
} else {
    echo "$number is not an Armstrong Number";
}
?>
```

OUTPUT:-

```
Enter A Number: 153
Yes, 153 is an Armstrong Number
```

II. Reverse a Number.

CODE:-

```
<?php
$number = readline("Enter a number: ");
$reverseNumber = 0;
while ($number > 0) {
    $digit = $number % 10;
    $reverseNumber = $reverseNumber * 10 + $digit;
    $number = (int)($number / 10);
}
echo "Reversed number: $reverseNumber";
```

OUTPUT:-

```
Enter a Number: 134
Reversed number: 431
```

III. Check Whether a number is a Prime number or not.

CODE:-

```
<?php
function check_prime($num)
{
    if ($num == 1)
        return 0;
    for ($i = 2; $i <= $num/2; $i++)
    {
        if ($num % $i == 0)
            return 0;
    }
    return 1;
}
$num = readline("Enter Your Choice: ");
$flag_val = check_prime($num);
if ($flag_val == 1)
    echo "It is a prime number";
else
    echo "It is a non-prime number"
?>
```

OUTPUT:-

Enter Your Choice: 5

It is a Prime number

IV. Check Whether a number is Palindrome or not.

CODE:-

```
<?

$input = readline("Enter Your Choice: ");

$number = $input;
```

```

$sum = 0;

while ((int)$number > 0) {

    $rem = $number % 10;

    $sum = $sum * 10 + $rem;

    $number = (int)($number / 10);

}

if ($input == $sum) {

    echo "$input is a Palindrome number";

} else {

    echo "$input is not a Palindrome";

}

?>

```

OUTPUT:-

Enter Your Choice: 12321

12321 is a Palindrome number

V. Print Star(*) Pattern.

CODE:-

```

<?php
$s = readline("Enter Your Choice: ");
for ($i = 0; $i <= $s; $i++) {
    for ($j = $s - $i; $j >= 1; $j--) {
        echo "* ";
    }
    echo "\n";
}

?>

```

OUTPUT:-

Enter Your Choice: 5

* * * * *

* * * *

* * *

* *

*

VI. Write a PHP program to demonstrate the mixing of decisions and looping states with HTML script.

CODE:-

```
<html>
<head>
    <title>Simple Calculator</title>
</head>
<body>
    <h1>Simple Calculator</h1>
    <form method="post">
        <input type="number" name="num1" placeholder="Enter first
            number" required>
        <br><br>
        <input type="number" name="num2" placeholder="Enter second
            number" required>
        <br><br>
        <select name="operation">
            <option value="add">+</option>
            <option value="subtract">-</option>
            <option value="multiply">*</option>
            <option value="divide">/</option>
        </select>
        <input type="submit" value="Calculate">
    </form>

    <?php
    if (isset($_POST['num1']) && isset($_POST['num2'])) {
        $num1 = $_POST['num1'];
```

```

$num2 = $_POST['num2'];
$operation = $_POST['operation'];
$result = 0;

switch ($operation) {
    case 'add':
        $result = $num1 + $num2;
        break;
    case 'subtract':
        $result = $num1 - $num2;
        break;
    case 'multiply':
        $result = $num1 * $num2;
        break;
    case 'divide':
        if ($num2 != 0) {
            $result = $num1 / $num2;
        } else {
            echo "<p class='error'>Error: Division by zero.</p>";
        }
        break;
    default:
        echo "<p class='error'>Error: Invalid operation.</p>";
        break;
}

echo "<p>Result: $num1 $operation $num2 = $result</p>";
}
?>
</body>
</html>

```